

AI for Communication and Marketing

Lab Exam – part 1

Academic Year: 2025/2026

Course: B.Sc. in Artificial Intelligence

1. Business Scenario

You have been hired as a **Junior Data Scientist** by a retail company. The Marketing Department has collected two years' worth of data on their customers but lacks the analytical framework to use it effectively.

Your goal is to perform a complete end-to-end analysis - from assessing data integrity to identifying high-value customer segments - to drive a data-driven communication strategy.

2. Dataset

You will use the **marketing_campaign.csv** dataset:

Column	Description
ID	Customer's unique identifier
Year_Birth	Customer's birth year
Education	Customer's education level
Marital_Status	Customer's marital status
Income	Customer's yearly household income
Kidhome	Number of small children in customer's household
Teenhome	Number of teenagers in customer's household
Dt_Customer	Date of customer's enrollment with the company
Recency	Number of days since customer's last purchase
MntWines	Amount spent on wine in last 2 years
MntFruits	Amount spent on fruits in last 2 years
MntMeatProducts	Amount spent on meat in last 2 years
MntFishProducts	Amount spent on fish in last 2 years
MntSweetProducts	Amount spent on sweets in last 2 years
MntGoldProds	Amount spent on gold in last 2 years
NumDealsPurchases	Number of purchases made with a discount
NumWebPurchases	Number of purchases made through the company's web site
NumCatalogPurchases	Number of purchases made using a catalogue
NumStorePurchases	Number of purchases made directly in stores
NumWebVisitsMonth	Number of visits to company's web site in the last month
AcceptedCmp3	1 if customer accepted the offer in the 3rd campaign, 0 otherwise
AcceptedCmp...	1 if customer accepted the offer in the ... campaign, 0 otherwise
AcceptedCmp1	1 if customer accepted the offer in the 1st campaign, 0 otherwise
Complain	1 if customer complained in the last 2 years, 0 otherwise
Response	1 if customer accepted the offer in the last campaign, 0 otherwise

3. Exam Requirements

Your submission must consist of two parts:

1. **A Jupyter Notebook (.ipynb):** Containing clean, commented Python code
2. **A Presentation (2-3 Slides .pdf format):** Summarizing key findings and marketing recommendations for non-technical stakeholders

Task A: Data Audit & Quality

- Perform a Data Audit: identify missing values, duplicates and data type inconsistencies
- Identify and handle Outliers
- Document your cleaning process: Why did you choose to drop, impute, or transform specific data points?

Task B: Exploratory Data Analysis (EDA)

- Analyze the correlation between socio-demographic features and total spending
- Visualize the distribution of purchases across different channels
- **Communication Insight:** Identify one "hidden" trend in the data that could be used for a targeted promotional campaign

Task C: RFM Calculation & Heuristic Segmentation

- Engineer the Recency, Frequency, and Monetary features
- Assign scores to customers based on quintiles
- Define at least 4 segments based on these scores

Task D: Unsupervised Learning (Cluster Analysis)

- Apply the K-Means algorithm to segment customers
- Justify the optimal number of clusters using the Elbow Method or Silhouette Score
- Compare the Clusters with your RFM segments. Do they overlap? What additional information do the clusters capture that RFM might have missed?

Task E: Communication Strategy

- Based on your findings, suggest a specific Communication Channel and Message Tone for your identified segments